

Incremental Novel Class Discovery Experiments

Metrics & Terminology

1. Known-Class Accuracy (KCA):
$$KCA = (\text{Correct predictions on known classes}) / (\text{Total samples of known classes})$$
2. Forgetting = (Best past accuracy – Current accuracy), averaged over all learned classes
Example:
If dog accuracy was 0.76 earlier and becomes 0.21 later:
→ Forgetting = $0.76 - 0.21 = 0.55$
3. Average Forgetting = mean forgetting over all learned classes

Experiment set 1:

All experiments in set 1 followed cross entropy loss during task0 and no contrastive learning.

Experiment 1.1: Single Novel Class (Dog)

Setup:

Task 0: Supervised training on airplane and automobile (10,000 images).

Tasks 1–5: Stream of unseen class (dog), 1000 images per task, no labels.

Model performs novelty detection, buffering, clustering (HDBSCAN), and promotion.

Task-wise Metrics

Task	KCA	Avg. Forgetting	Event
0	0.988	0.000	Initialisation
1	0.988	0.000	Buffering
2	0.988	0.000	Buffering
3	0.866	0.093	Dog promoted

4	0.866	0.062	Stable
5	0.866	0.062	Stable

Class-wise Accuracy Trend

Task	airplane	automobile	dog
0	0.989	0.987	-
1	0.989	0.987	-
2	0.989	0.987	-
3	0.885	0.905	0.808
4	0.885	0.905	0.808
5	0.885	0.905	0.808

Purity Table

Task	ClusterID	Dominant Class	Purity	Cluster Size	Status
3	0	dog	1.00	192	Promoted

Observation:

The model handles one new class quite well because the data is simple and comes in gradually. Since there is no confusion with other new classes, clustering works cleanly and the old classes are mostly preserved. The drop is simply because of realignment of the boundaries.

Experiment 1.2: Sequential Classes – Partial Open World

Setup

Task 0: Supervised initialization.

Tasks 1–40: Sequential classes (5 tasks each, 1000 images/task, no labels).

Order: bird → cat → deer → dog → frog → horse → ship → truck.

Buffer cleared after each class phase.

Key Metrics

Task	KCA	Avg. Forgetting	Event
0-17	0.988	0.000	No promotion
18	0.863	0.076	Dog promotion
21	0.714	0.256	Frog promotion
26	0.725	0.202	Horse promotion
31	0.739	0.166	Ship promotion
37	0.729	0.128	Truck promotion

Class-wise Accuracy Trend

Task	airplane	automobile	dog	frog	horse	ship	truck
0	0.989	0.987	-	-	-	-	-
18	0.907	0.916	0.766	-	-	-	-
21	0.860	0.898	0.215	0.883	-	-	-
26	0.799	0.904	0.249	0.865	0.809	-	-
31	0.733	0.882	0.375	0.838	0.774	0.831	-
37	0.682	0.862	0.497	0.859	0.751	0.845	0.604

Purity Tables

Promoted Clusters

Class	Task	ClusterID	Purity	Size
dog	18	0	1.00	192
frog	21	0	1.00	250

horse	26	2	1.00	224
ship	31	1	1.00	319
truck	37	0	1.00	535

Observation

Even though each new class is learned correctly, the model slowly forgets older classes over time. This happens because every new class slightly changes the internal representation, which affects what was learned earlier.

Experiment 1.3: Mixed Open-World Setting

1.3a: (All Classes together)

Setup

Task 0: Supervised initialization.

Tasks 1–40: All 10 classes mixed (~1250 samples/task), no labels.

Key Metrics

Task	KCA	Avg. Forgetting	Event
0	0.988	0.000	Initialization
2	0.896	0.082	Frog promotion
3	0.713	0.249	Deer promotion
4	0.530	0.402	Bird and Truck both promoted

Class-wise Accuracy Trend

Task	airplane	automobile	frog	deer	bird	truck
0	0.989	0.987	-	-	-	-
2	0.894	0.918	0.875	-	-	-

3	0.735	0.908	0.462	0.747	-	-
4	0.710	0.749	0.230	0.300	0.493	0.701

Purity Table

Class	Task	ClusterID	Purity	Size
frog	2	1	0.17	166
deer	3	1	0.17	345
bird	4	0	0.15	321
Truck	4	1	0.32	152

Observation

Here everything breaks because all classes are mixed together from the start. The model is not able to form clean clusters, so it ends up learning incorrect groupings and overall performance drops a lot.

1.3b: (Modified setting – 2-3 Classes together at a time)

Stream setup 1:

In this method, new classes are completely exhausted (all 5,000 images shown) within their specific phase.

After their phase ends, they become "reminders" (100 reused images per task) alongside Task 0 classes.

Task 0 (Initialization)

Class 0 (Airplane): 5,000 images

Class 1 (Automobile): 5,000 images

Total Batch Size: 10,000

Phase 1 (Tasks 1 to 13) - Exhausting Class 2 & Class 3

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (New Wave): ~384 images each (New) (5,000 imgs / 13 tasks)

Total Batch Size per Task: ~968 images

Phase 2 (Tasks 14 to 26) - Exhausting Class 4, Class 5, Class 6

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (Reminders): 100 images each (Reused from Phase 1)

Class 4, Class 5, Class 6 (New Wave): ~384 images each (New) (5,000 imgs / 13 tasks)

Total Batch Size per Task: ~1,552 images

Phase 3 (Tasks 27 to 39) - Exhausting Class 7, Class 8, Class 9

Class 0 to Class 6 (Reminders): 100 images each (Reused) -> 700 images total

Class 7, Class 8, Class 9 (New Wave): ~384 images each (New) (5,000 imgs / 13 tasks)

Total Batch Size per Task: ~1,852 images

Results:

Key Metrics

Task	KCA	Avg. Forgetting	Event
0-4	0.992	0.000	Initialisation
5	0.917	0.078	Cat promotion
6	0.706	0.222	Bird promotion
15	0.509	0.345	Deer & Frog promotion
27	0.489	0.278	Truck & Ship promotion
39	0.489	0.208	End of stream

Class-wise Accuracy Trend

Task	airplane	automobile	cat	bird	deer	frog	truck	ship
0	0.994	0.989	-	-	-	-	-	-
5	0.863	0.964	0.923	-	-	-	-	-
6	0.843	0.945	0.452	0.583	-	-	-	-
15	0.596	0.951	0.102	0.462	0.241	0.705	-	-
27	0.539	0.894	0.288	0.440	0.098	0.510	0.554	0.588
39	0.539	0.894	0.288	0.440	0.098	0.510	0.554	0.588

Purity Table

Class	Task	ClusterID	Purity	Size
cat	5	0	0.60	1832
bird	6	2	0.54	292
deer	15	0	0.41	263
frog	15	2	0.51	207

truck	27	0	0.60	205
ship	27	1	0.37	284

Stream setup 2:

In this method, we do not exhaust the new classes immediately. Instead, we take their 5,000 images and spread them evenly across all remaining tasks until Task 39.

Task 0 (Initialization)

Class 0 (Airplane): 5,000 images

Class 1 (Automobile): 5,000 images

Total Batch Size: 10,000

Phase 1 (Tasks 1 to 13) - Starting Class 2 & Class 3

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (New Stream): ~128 images each (New) (5,000 imgs spread over 39 tasks)

Total Batch Size per Task: ~456 images

Phase 2 (Tasks 14 to 26) - Starting Class 4, Class 5, Class 6

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (Continuing Stream): ~128 images each (New)

Class 4, Class 5, Class 6 (New Stream): ~192 images each (New) (5,000 imgs spread over 26 tasks)

Total Batch Size per Task: ~1,032 images

Phase 3 (Tasks 27 to 39) - Starting Class 7, Class 8, Class 9

Class 0, Class 1 (Reminders): 100 images each (Reused)

Class 2, Class 3 (Continuing Stream): ~128 images each (New)

Class 4, Class 5, Class 6 (Continuing Stream): ~192 images each (New)

Class 7, Class 8, Class 9 (New Stream): ~384 images each (New) (5,000 imgs spread over 13 tasks)

Total Batch Size per Task: ~2,184 images

Results

Key Metrics

Task	KCA	Avg. Forgetting	Event
0-1	0.992	0.000	Initialisation
2	0.902	0.078	Cat promotion
4	0.681	0.194	Bird promotion
18	0.603	0.206	Frog promotion
27	0.573	0.197	Truck promotion
33	0.518	0.218	Ship promotion
39	0.518	0.187	End of stream

Class-wise Accuracy Trend

Task	airplane	automobile	cat	bird	frog	truck	ship
0	0.994	0.989	-	-	-	-	-
2	0.883	0.945	0.877	-	-	-	-
4	0.792	0.949	0.536	0.449	-	-	-
18	0.842	0.897	0.448	0.297	0.532	-	-
27	0.792	0.831	0.464	0.374	0.394	0.585	-
33	0.788	0.813	0.277	0.402	0.308	0.529	0.510
39	0.788	0.813	0.277	0.402	0.308	0.529	0.510

Purity Table

Class	Task	ClusterID	Purity	Size
cat	2	1	0.53	318
bird	4	1	0.39	212
frog	18	0	0.35	241
truck	27	1	0.57	229
ship	33	0	0.27	278

Experiment set 2:

1. Experiments use cross-entropy loss and margin contrastive loss, applied during Task 0 for contrastive learning.
2. Joint Training with promoted samples (upper bound) refers to a fully supervised setting in which all available data, both the original base class samples and the samples promoted by the algorithm are jointly used for training.

Experiment 2.1: Single Novel Class (Dog)

Setup

Task 0: Supervised training on airplane and automobile (10,000 images).

Tasks 1–5: Stream of unseen class (dog), 1000 images per task, no labels.

Model performs novelty detection, buffering, clustering (HDBSCAN), and promotion.

Experiment 2.1a: Joint Training with Promoted Samples - Upper Bound

🏆 JOINT UPPER-BOUND SUMMARY (Static Data, No Forgetting)					
Method	Dog Samples	Airplane Acc	Automobile Acc	Dog Acc	Overall Acc
baseline_ce	192	0.991	0.991	0.743	0.908
soft_center_loss	176	0.981	0.993	0.744	0.906
margin_contrastive	518	0.985	0.990	0.837	0.937
triplet	282	0.988	0.990	0.792	0.923

Experiment 2.1b: Incremental Setup Key Metrics

Method	Dog Samples	Airplane Acc	Automobile Acc	Dog Acc	Final KCA	Avg Forgetting	Final Classes
baseline_ce	192	0.908	0.918	0.759	0.862	0.050	3
soft_center_loss	176	0.944	0.130	0.794	0.623	0.300	3
margin_contrastive	518	0.862	0.875	0.975	0.904	0.081	3
triplet	282	0.957	0.956	0.860	0.924	0.021	3

Execution completely finished. t-SNE Plots saved inside respective method folders.

Experiment 2.2: Partial Open World - Sequential Classes 5 tasks each

Setup

Task 0: Supervised initialization.

Tasks 1–40: Sequential classes (5 tasks each, 1000 images/task, no labels).

Order: bird → cat → deer → dog → frog → horse → ship → truck.

Buffer cleared after each class phase.

Key Metrics:

Final KCA and Average Forgetting for Different Techniques Under Varying Hyperparameter Configurations

Global Hyperparameter Grid Search (41-Task Gauntlet)

HPT	Method	Final KCA	Avg Forg	Classes Learned	Cls0	Cls1	Cls2	Cls3	Cls4	Cls5	Cls6	Cls7	Cls8	Cls9
Strict	baseline_ce	0.612	0.237	9	0.56	0.80	0.44	0.32	0.37	0.68	0.82	0.65	0.88	0.00
Strict	soft_center_loss	0.582	0.298	8	0.13	0.71	0.38	0.28	0.00	0.63	0.77	0.00	0.92	0.84
Strict	margin_contrastive	0.622	0.214	10	0.58	0.73	0.48	0.28	0.40	0.60	0.78	0.75	0.80	0.82
Strict	triplet	0.491	0.252	10	0.65	0.75	0.42	0.39	0.32	0.39	0.73	0.44	0.20	0.62
Moderate	baseline_ce	0.612	0.237	9	0.56	0.80	0.44	0.32	0.37	0.68	0.82	0.65	0.88	0.00
Moderate	soft_center_loss	0.605	0.243	9	0.56	0.67	0.35	0.33	0.48	0.61	0.78	0.00	0.81	0.85
Moderate	margin_contrastive	0.617	0.226	10	0.48	0.68	0.47	0.37	0.40	0.61	0.76	0.73	0.83	0.83
Moderate	triplet	0.491	0.252	10	0.65	0.75	0.42	0.39	0.32	0.39	0.73	0.44	0.20	0.62
Relaxed	baseline_ce	0.430	0.362	10	0.00	0.41	0.40	0.30	0.33	0.65	0.78	0.73	0.30	0.40
Relaxed	soft_center_loss	0.511	0.322	9	0.36	0.68	0.42	0.28	0.42	0.65	0.72	0.00	0.27	0.80
Relaxed	margin_contrastive	0.573	0.380	5	0.00	0.73	0.45	0.00	0.00	0.00	0.00	0.00	0.85	0.83
Relaxed	triplet	0.428	0.287	10	0.09	0.74	0.43	0.35	0.38	0.52	0.66	0.36	0.20	0.57

Optimization of the Above Strict Hyperparameter Configuration for Margin Contrastive Loss

📌 [UPPER BOUND] Took 40 mins to run on a 10-class stream! I added the novelty filter for the real class.

📌 [METRIC] Known-Class Acc: 0.685 | Avg Forgetting: 0.185

- Class 'airplane' Acc: 0.644
- Class 'automobile' Acc: 0.873
- Class 'bird' Acc: 0.548
- Class 'cat' Acc: 0.345
- Class 'deer' Acc: 0.675
- Class 'dog' Acc: 0.448
- Class 'frog' Acc: 0.832
- Class 'horse' Acc: 0.905
- Class 'ship' Acc: 0.799
- Class 'truck' Acc: 0.781

📊 FINAL OPTIMIZED RUN SUMMARY (10-Class Partial Open World)

HPT	Method	Final KCA	Avg Forg	Classes Learned	Cls0	Cls1	Cls2	Cls3	Cls4	Cls5	Cls6	Cls7	Cls8	Cls9
Strict (Optimized)	margin_contrastive	0.685	0.185	10	0.64	0.87	0.55	0.34	0.68	0.45	0.83	0.91	0.80	0.78

✅ Script complete. Descriptive table saved to 'final_optimized_margincontrastive_stricthpt_summary_table.png'

Final KCA and Average Forgetting for Margin Contrastive Loss with Optimized Strict Hyperparameters (Joint Training with Promoted Samples - Upper Bound)

📌 Fetching Base Datasets (Airplanes & Autos)...

- Base Data Loaded: 10000 images (Classes 0 & 1)
- Loaded 'cat' (Class 3): 254 validated images.
- Loaded 'ship' (Class 8): 242 validated images.
- Loaded 'dog' (Class 5): 221 validated images.
- Loaded 'deer' (Class 4): 1174 validated images.
- Loaded 'bird' (Class 2): 265 validated images.
- Loaded 'frog' (Class 6): 379 validated images.
- Loaded 'truck' (Class 9): 281 validated images.
- Loaded 'horse' (Class 7): 1903 validated images.

📊 Final Joint Dataset Size: 14719 images across 10 classes.

📌 Training static Upper Bound model on exact CL data subset...

- Epoch 5/30 | Loss: 0.7291
- Epoch 10/30 | Loss: 0.3136
- Epoch 15/30 | Loss: 0.2048
- Epoch 20/30 | Loss: 0.1342
- Epoch 25/30 | Loss: 0.1030
- Epoch 30/30 | Loss: 0.0818

📌 ALGORITHM-SPECIFIC JOINT UPPER-BOUND SUMMARY (Zero Forgetting on Discovered Data)

Method	UB-KCA	Cls0_air	Cls1_aut	Cls2_bir	Cls3_cat	Cls4_dee	Cls5_dog	Cls6_fro	Cls7_hor	Cls8_shi	Cls9_tru
margin_contrastive	0.758	0.979	0.983	0.555	0.447	0.880	0.591	0.837	0.919	0.659	0.728

Experiment 2.3: Open World (mixing of 2-3 classes in 13 wave tasks)

Experiment 2.3a
(Applying the Best Margin Contrastive Setup from Partial Open World
(Final KCA: 68.5)):

FINAL OPTIMIZED RUN SUMMARY (Multi-Class Wave Stream)

HPT	Method	Final KCA	Avg Forg	Classes Learned	Cls0	Cls1	Cls2	Cls3	Cls4	Cls5	Cls6	Cls7	Cls8	Cls9
Strict (Optimized)	margin_contrastive	0.538	0.244	7	0.65	0.76	0.00	0.45	0.00	0.00	0.32	0.22	0.68	0.69

Script complete. Descriptive table saved to 'final_optimized_margincontrastive_strictcpt_summary_table.png'

Class	Task	Cluster ID	Purity	Size
Cat	1	0	0.56	318
Frog	14	0	0.48	212
Horse	27	1	0.18	241
Ship	28	0	0.42	229
Truck	28	1	0.67	278

Experiment 2.3b
(Purification Methods):

MASTER ABLATION SUMMARY: TRACK A (CLUSTER PURIFICATION)

Experiment	Final KCA	Avg Forg	Classes	Init Pur	Core Pur	C0(air)	C1(aut)	C2(bir)	C3(cat)	C4(dee)	C5(dog)	C6(fro)	C7(hor)	C8(shi)	C9(tru)
Baseline (Earlier Code)	0.538	0.244	7	0.46	0.46	0.65	0.76	0.00	0.45	0.00	0.00	0.32	0.22	0.68	0.69
Centroid Distance (Top 50%)	0.506	0.293	5	0.45	0.50	0.21	0.91	0.00	0.47	0.19	0.00	0.75	0.00	0.00	0.00
Known Repulsion (< 0.50)	0.991	0.000	2	0.00	0.00	0.98	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
HDBSCAN Prob (>= 0.80)	0.415	0.440	5	0.42	0.47	0.00	0.90	0.00	0.20	0.00	0.00	0.33	0.64	0.00	0.00
Entropy / MSP (< 0.85)	0.789	0.201	3	0.50	0.73	0.47	0.91	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00
k-NN Memory Overlap (k=5, max=2)	0.991	0.000	2	0.00	0.00	0.98	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Ablation complete. Master table saved to 'ablation_track_A_summary.png'

MASTER ABLATION SUMMARY: TRACK A (PHASE 2)

Experiment	Final KCA	Avg Forg	Classes	Init Pur	Core Pur	C0(air)	C1(aut)	C2(bir)	C3(cat)	C4(dee)	C5(dog)	C6(fro)	C7(hor)	C8(shi)	C9(tru)
Baseline (Control)	0.538	0.244	7	0.46	0.46	0.65	0.76	0.00	0.45	0.00	0.00	0.32	0.22	0.68	0.69
MSP Tight (< 0.85)	0.789	0.201	3	0.50	0.73	0.47	0.91	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00
MSP Medium (< 0.95)	0.406	0.559	5	0.38	0.42	0.07	0.74	0.00	0.16	0.00	0.00	0.00	0.90	0.00	0.57
Centroid (Top 75%)	0.647	0.338	3	0.56	0.59	0.00	0.97	0.00	0.97	0.00	0.00	0.00	0.00	0.00	0.00
Known Repulsion (< 0.55)	0.792	0.198	3	0.59	0.65	0.52	0.87	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00
HDBSCAN Prob (>= 0.40)	0.510	0.263	7	0.47	0.47	0.55	0.75	0.00	0.37	0.00	0.00	0.34	0.28	0.69	0.61
k-NN (k=7, overlap=6)	0.991	0.000	2	0.00	0.00	0.98	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Ablation complete. Master table saved to 'ablation_track_A_phase2_summary.png'

(Clustering techniques grid search):

MASTER ABLATION SUMMARY: TRACK B (COSINE CLUSTERING)														
Experiment	Final KCA	Avg Forg	Classes	Raw Pur	C0(air)	C1(aut)	C2(bin)	C3(cat)	C4(dee)	C5(dog)	C6(fro)	C7(hor)	C8(shi)	C9(tru)
Baseline (HDSCAN Euclidean)	0.538	0.244	7	0.46	0.65	0.76	0.00	0.45	0.00	0.00	0.32	0.22	0.68	0.69
HDSCAN Cosine	0.538	0.244	7	0.46	0.65	0.76	0.00	0.45	0.00	0.00	0.32	0.22	0.68	0.69
Spherical K-Means (Dynamic)	0.390	0.346	8	0.33	0.39	0.78	0.00	0.21	0.18	0.33	0.32	0.45	0.00	0.46
FINCH 1-NN Graph (Cosine)	0.726	0.264	3	0.58	0.24	0.94	0.00	0.99	0.00	0.00	0.00	0.00	0.00	0.00
SHN-DBSCAN (Shared NN)	0.480	0.382	6	0.45	0.32	0.66	0.00	0.41	0.46	0.00	0.34	0.00	0.00	0.09
Agglomerative (Cosine + Avg)	0.576	0.381	6	0.53	0.46	0.77	0.00	0.23	0.00	0.56	0.00	0.00	0.79	0.63
✔ Ablation complete. Master table saved to 'ablation_track_b_summary.png'														

(Open Space Losses):

MASTER ABLATION SUMMARY: TRACK C (OPEN-SPACE LOSS)														
Experiment	Final KCA	Avg Forg	Classes	Raw Pur	C0(air)	C1(aut)	C2(blir)	C3(cat)	C4(dee)	C5(dog)	C6(fro)	C7(hor)	C8(shl)	C9(tru)
Baseline (No Space Curving)	0.530	0.244	7	0.46	0.65	0.76	0.00	0.45	0.00	0.00	0.22	0.22	0.68	0.69
Background Repulsion Loss	0.519	0.286	8	0.43	0.61	0.82	0.00	0.38	0.41	0.00	0.43	0.48	0.69	0.50
Virtual Placeholder (K+1)	0.446	0.311	6	0.45	0.29	0.85	0.00	0.52	0.46	0.16	0.39	0.00	0.00	0.00
Entropic Open-Space Loss	0.427	0.362	9	0.48	0.52	0.73	0.48	0.00	0.24	0.36	0.37	0.00	0.66	0.48
Ablation complete. Master table saved to 'ablation_track_C_summary.png'														

MASTER ABLATION SUMMARY: TRACK C (ADVANCED SOTA LOSS)

Experiment	Final KCA	Avg Forg	Classes	Raw Pur	C0(air)	C1(aut)	C2(bir)	C3(cat)	C4(dee)	C5(dog)	C6(fro)	C7(hor)	C8(sh1)	C9(tru)
Baseline (No Space Carving)	0.538	0.244	7	0.46	0.65	0.76	0.00	0.45	0.00	0.00	0.32	0.22	0.68	0.69
Objectosphere (Mag Collapse)	0.432	0.350	8	0.45	0.45	0.75	0.00	0.20	0.12	0.00	0.55	0.91	0.03	0.44
EDL (Dirichlet Evidence)	0.464	0.337	8	0.42	0.43	0.82	0.00	0.23	0.22	0.00	0.38	0.37	0.73	0.54
Proxy Reciprocal Points (RPL)	0.476	0.363	8	0.45	0.73	0.81	0.00	0.10	0.00	0.20	0.36	0.51	0.56	0.55

Ablation complete. Master table saved to 'ablation_track_C_advanced_summary.png'

Combining Experiments 2.3b, 2.3c, and 2.3d for Improved Results:

MASTER ABLATION SUMMARY: FINAL VOLTRON PIPELINE

Experiment	Final KCA	Avg Forg	Classes	Raw Pur	C0(air)	C1(aut)	C2(bir)	C3(cat)	C4(dee)	C5(dog)	C6(fro)	C7(hor)	C8(shi)	C9(tru)
The Voltron Pipeline (Agglo + Entropy + Lwf + Hard Lock)	0.379	0.279	10	0.45	0.78	0.75	0.27	0.02	0.05	0.06	0.28	0.34	0.58	0.65

✓ Ablation complete. Master table saved to 'ablation_final_voltron_summary.png'

Experiment 2.3e

(Best Results So Far with Contrastive Warmup, MNN Graph, and Candidate Stash):

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FINAL OPTIMIZED RUN SUMMARY (K-Means Anchor + Max Discovery Gates)														
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HPT	Method	Final KCA	Avg Forg	Classes Learned	Cls0	Cls1	Cls2	Cls3	Cls4	Cls5	Cls6	Cls7	Cls8	Cls9

The Final Synthesis	margin_contrastive	0.602	0.306	7	0.56	0.61	0.00	0.64	0.53	0.00	0.41	0.00	0.76	0.72
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✔ Script complete.														

Class	Task	Stash Purity	Final Stash Size
cat	1	0.62	385
deer	14	0.36	346
frog	16	0.38	417
ship	27	0.54	385
truck	28	0.49	385